



Virtual Speech-Language Therapy for Individuals with Communication Disorders: Current Evidence, Limitations, and Benefits

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Abstract

Purpose of Review To review currently available research on virtual therapy for individuals with speech-language disorders. Virtual speech-language therapy refers to speech-language therapy delivered via computer-simulation technology to improve speech and language abilities for children and adults with communication disorders. In addition, this paper addressed limitations of previous studies and benefits of virtual therapy.

Recent Findings Recent research reports suggest that virtual therapy is a viable option for treating children and adults with speech-language disorders. However, most current research studies were conducted without a rigorous experimental design. Also, the quality of technology is indeterminate and application to real clinical practice is currently lacking. More studies should be conducted with a rigorous experimental design as well as advanced technology in the future.

Summary This paper has reviewed current evidence related to the effectiveness of virtual speech-language intervention for children and adults with speech-language impairment. This paper discussed limitations of the current research and benefits of virtual therapy.

Keywords Virtual therapy · Speech-language intervention · Virtual therapist · Virtual reality · Augmented reality

Introduction

Definition and Scope of this Review

Virtual refers to “being on or simulated on a computer or computer network” [1]. Virtual speech-language therapy refers to speech-language therapy delivered via computer simulation technology to improve communication abilities for individuals with communication disorders. Virtual speech or language therapy may be referred to as a broader concept such as speech or language therapy via interactive computer technology [2•]. However, if virtual therapy includes assessment and treatment utilizing any types of computer hardware and/or software programs, it may result in being very heterogeneous.

Thus, I will define the term “virtual” specifically to refer to the use of a simulated environment or a therapist for intervention. Based on the definition, speech-language intervention utilizing a variety of computer software to improve communication or behavioral skills may be referred to as computer-assisted or technology-assisted intervention. For example, therapy provided with software programs designed to improve phonological awareness, vocabulary, grammar, or reading comprehension may be called a computer-assisted speech-language intervention. Research using computer-assisted technology was not included in this review since that is not the scope of this review.

Another computer technology that is being rapidly adopted by healthcare professions is telehealth. Telehealth is often substituted by various terminologies such as e-health, telemedicine, telerehabilitation, or telepractice depending on professions or fields. In general, telehealth refers to the provision of healthcare services such as assessment, intervention, and consultation via telecommunication technologies to patients residing at a distance from any location where they could not receive direct healthcare services [3]. Telehealth is a kind of online or electronic learning, which is defined as education

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that takes place over the internet across distance. Online or electronic learning could be distinguished from telehealth in that the service recipients are mainly individuals without disabilities. Also, the various terminologies that fall under the umbrella term “telehealth” are delineated by the type of service provided by a healthcare profession (e.g., telemedicine for medical service or telepractice for speech-language pathology service). Although telehealth also utilizes computer technologies, telehealth is distinguished from virtual therapy because telehealth typically takes place in a real, not simulated, environment and is provided by a real therapist either synchronously or asynchronously. It should be noted that, however, attempts to combine telehealth with a virtual environment or therapist have been emerging [4]. Thus, only telehealth research incorporating virtual therapy was included in this review.

Virtual therapy has been adopted in various healthcare fields such as medical rehabilitation [5, 6], physical therapy [7], mental health [8], medical diagnosis [9, 10], and surgery training [11, 12]. The current review only included studies where virtual therapy was implemented within the scope of practice for a speech-language pathology. Any research investigating the effect of virtual therapy on motor, hearing, socio-emotional aspects, and daily-living skills, of adults and children with disorders was not addressed either.

Finally, although a number of manuscripts were available that report on the positive effect of virtual therapy in adults and children with communication disorders, some of the evidence from these studies was not included in the current review because the studies mainly focused on providing a description of technical developments and the clinical application of virtual therapy was mainly qualitatively described with no specific experimental procedures. Thus, the current review included research papers with a substantial number of participants with descriptions of the details of experimental procedures.

The primary goal of this paper, therefore, was to review how virtual therapy using computer-generated simulation has been used in speech-language intervention for children and adults with communication disorders.

Current Findings on Virtual Therapy for Adults with Communication Disorders

In spite of the advancement of virtual therapy in the medical and other healthcare fields (e.g., physical therapy), until recently, only a few attempts have been made to provide speech-language services via virtual therapy for individuals with communication disorders even though virtual therapy has great potential to enhance communication. Most research utilizing virtual therapy in adult patients has been conducted with individuals with aphasia. In addition, limited studies are

available with adult motor speech disorders, voice disorders, and stuttering.

A research group developed a virtual computer program and conducted a series of studies in which a virtual agent taught various language skills to adult patients with neurogenic communication disorders. In an initial study, Cherney, Halper, Holland, and Cole [13•] evaluated the feasibility of a virtual therapist to teach participants with aphasia. Qualitative reports showed that all three participants improved their spontaneous speech, comprehension, repetition, and naming skills. In a further study, Thompson, Choy, Holland, and Cole [14] found positive results in six patients with aphasia when a virtual clinician presented a sentence task (e.g., the sentence-picture naming task) to patients. They also found no significant differences between acquisitions or generalization patterns as compared with the results of a human clinician-delivered intervention. Lee, Kaye, and Cherney [15] also used the same software program to investigate the relationship between amount of treatment and improvement of conversational performance in patients with non-fluent aphasia. Seventeen participants practiced for 3 weeks to measure percent change in content words and rate. The authors found that amount of treatment was significantly correlated with improvement in content in participants with more severe aphasia and significantly correlated with improvement in rate among participants with less severe aphasia. Later, Vuuren, and Cherney [16••] delivered virtual speech and language therapy using a similar program to eight patients with aphasia. The authors reported that the overall effect size for the virtual treatment was large and similar to what would be expected when treatment is delivered by a clinician in the face-to-face sessions. A similar virtual program was also used for voice training in Parkinson's disease [17]. The well-established LSVT (Lee Silverman Voice Treatment) was implemented by a virtual therapist to a patient. As a pilot study, all findings were reported descriptively. Main findings were obtained through an interview with the participant who showed a positive experience with the virtual therapist.

Another group of researchers [18••] also investigated the effect of virtual clinicians in virtual therapy with patients with aphasia. Four patients practiced dialogs with a human clinician and a virtual clinician. Dialogs before and after virtual and face-to-face sessions were compared. It was found that some of the patients with aphasia produced more verbal outputs with the virtual clinician while some patients showed no difference between the two conditions. The authors claimed that these results provide support for the feasibility and applicability of virtual technology to real-life communication contexts to improve functional communication in patients with aphasia.

Abad et al. [19••] and Marshal et al. [20••] conducted studies where telehealth and virtual reality technology were combined. Unlike previous studies, these studies included a

relatively large number of participants (i.e., 16 and 20 participants, respectively). Abad et al. developed an online word-naming training program for patients with aphasia. Using automatic speech recognition technology, the program identifies patient's verbal responses. The authors found highly correlated word-naming scores and high-performance word verification rates between the program module and human evaluation, supporting the feasibility of virtual speech therapy. Marshal et al. developed an online virtual environment where each of participants is represented by a personalized avatar. Various communication abilities were assessed (e.g., functional communication, verbal fluency, word finding in conversation, words per minutes in narration). Besides achieving an excellent level of compliance during intervention, significant gains in functional communication was observed. However, no significant effects were found on the other outcome measures. Later, Amaya et al. [21] conducted a case study with two participants with aphasia using the same program used in Marshal et al. They found significant improvement in the naming of treated words; however, lexical gains were less evident.

Virtual therapy has been also used in stuttering assessment and intervention in adult populations. Brundage and colleagues [22, 23•, 24] conducted several studies using virtual reality technology with persons who stutter. They used virtual reality to simulate a job interview and public speaking situation. They found a positive correlation in stuttering severity between virtual therapy and a face-to-face interview as well as public speaking situations. The interviewer communication style affected the amount of stuttering with more stuttering during a challenging virtual interview. The authors argued that virtual environments would be useful for stuttering assessment and treatment because they cause affective, behavioral, and cognitive reactions in persons who stutter similar to what they encounter in the real world.

Current Findings on Virtual Therapy for Children with Communication Disorders

Research examining the effect of virtual therapy on speech-language issues in children has been mainly conducted in two areas. First, a virtual speech therapist or virtual 3-D talking head has been used in treating children with speech or articulatory disorders. Second, virtual reality or augmented reality has been utilized for treating children with autism spectrum disorders (ASD).

Massaro and Light [25•] investigated the efficacy of a computer-animated talking head and a virtual language tutor for speech perception and production in children with hearing loss. In the study, seven children with hearing loss aged 8 to 13 years received training concentrating on initial consonant voice/voiceless distinctions, consonant cluster distinctions, and fricative/affricate sound distinctions. The authors reported

that perception and production abilities significantly improved for all children from pre-test to post-test. They also found generalization to new words. However, speech production skills deteriorated after 6 weeks without training, suggesting that continuing practice was necessary to maintain the skills. Fagel and Madany [26] also developed a 3-D virtual talking head to help children with articulation disorders improve their /s/ and /z/ production. Eight children whose age ranged from 4 to 7 years received two sessions of speech therapy. They found that 6 out of the 8 children had a significant improvement following the first lesson and 3 out of the 8 children had persistent positive effect after the second learning session.

Saz et al. [27] evaluated a semi-automated virtual speech-language therapy system for children and young adults with dysarthria. This program was designed to facilitate phonation skills, phonetic articulation, and language-understanding skills of 14 patients aged 11–21 years. The accuracy of automatic speech recognition and pronunciation verification was mainly evaluated to validate the tool. The authors argued that the virtual speech therapy program delivers performance similar to experts' agreement rate. Schipor, Penitiuc, and Schipor [28] examined the possibility of automatic selection of exercises using a virtual system in treating 5- to 6-year-old children with /r/ and /s/ sound difficulties. Twenty children were divided into two groups. For the control group, all therapeutic decisions were made by the human speech therapist. For the experimental group, part of therapeutic decisions was made by the computer system. After 24 sessions, both groups of children showed similar production accuracy, suggesting that exercise choice by a computer system is feasible in a virtual speech intervention.

For children with ASD, virtual reality as well as augmented reality has been used to improve social, emotional, and communication abilities. While virtual reality completely replaces a real-world environment with a simulated one, augmented reality, one of the most common virtual reality technologies, is an interactive experience of a real-world environment where a real object is augmented by computer technologies in the real world. For example, a picture (e.g., bee) in a real world environment (e.g., book) flies when an augmented reality device scans the object. Chien, Chen, and Jeng [29] claimed that augmented reality may be particularly useful for teaching a subject where learners have difficulty experiencing the real world. Based on recent systematic reviews on the effectiveness of augmented and virtual reality for children and adolescents with ASD [30•, 31•], only a few studies examined communication abilities of children with ASD. And very limited research was conducted with rigorous experimental procedures.

Bernardini, Porayska-Pomsta, and Sampath [32] developed an intelligent virtual agent for helping children with ASD improve their social communication. The autonomous planning-based agent inhabits a virtual environment and acts

as an artificial social partner. Nineteen children with ASD participated in the study during which they played with the virtual agent for 10 to 20 min, several times a week over an 8-week period. The authors identified a positive trend, but the results showed that most of the children's outcome behaviors were not significantly improved before and after intervention using the virtual agent.

Parson [33••] developed a two-player virtual environment game where children had to verbally communicate and understand the perspective of the other child in order to be successful in completing the game. Six children with ASD aged 10–13 years and eight typically developed children aged 7–9 years participated in this study. The children played the games (1h total). The author found that the children with ASD made sustained endeavors to communicate with their partners during the virtual environment and were as reciprocal in their communication similar to typically developing children. In fact, the pairs of children with ASD made more communicative moves to their partners than the TD children. The author claimed that the 3-D virtual environments support more effective collaborative learning for children with ASD.

Bai, Blackwell, and Coulouris [34] examined whether AR stimulated the development of symbolic play for children with ASD. Twelve children ages 4–10 years old participated in both AR and non-AR play conditions. The children were asked to play with a block as an airplane, car, or truck. The authors found a significant increase in pretend play frequency and duration in the AR condition compared with the non-AR condition in 10 children. The other two children were not able to complete the study due to their joint attention issue. Most children and parents reported that AR provided a fun experience that allowed the children to become engaged in the play.

Finally, Chen and colleagues [35••, 36••] investigated how the 3-D facial models of virtual characters could be useful to teach children with ASD six basic facial expressions (happiness, sadness, fear, disgust, surprise, and anger). When teaching facial emotions using pictures, clinicians and patients often encounter difficulties because the patients lack engaging feelings. In their first study, Chen, Lee, and Lin [35••] tested three children with ASD age ranged 10–13 years in a single-subject multiple baseline design. After establishing a different baseline number for each child, the children were asked to imitate the visual feedback provided by the 3-D facial model to make facial expressions that correspond to the situation. The child's facial expressions were shown on the computer screen to help the children identify the right facial expression. An increasing trend on accurate responses of facial expressions was observed during the seven intervention sessions in all three children. All children maintained excellent facial recognition skills during the follow-up period. In another study, Chen, Lee, and Lin [36••] included six children with ASD age ranged from 11 to 13. Chen et al. adopted an augmented reality (AR)-based video modeling storybook (ARVMS) where

the children focused on a stable visual image printed in the storybook and then extended their attention to an animation of the story using a handheld device. After the three phases (baseline, intervention, and maintenance) test data was collected and the results showed that the ARVMS intervention provided an augmented visual indicator which had effectively attracted and maintained the attention of children with ASD to nonverbal social cues and helped them better understand the facial expressions and emotions of the storybook characters.

Limitations of the Current Research of Virtual Therapy

Although a number of attempts have been made to investigate whether virtual therapy is effective in treating children and adults with communication disorders, the current evidence may not provide strong evidence to show the effects of virtual therapy.

The major issue is that the previous studies were implemented without a rigorous experimental design. The majority of the studies were done as a case study. Among several experimental studies, they were conducted as a one group pre-test and post-test design or single-subject experimental design. Although these designs demonstrated positive effects on outcome behaviors, without a control group, it is difficult to claim how effective one treatment is as compared with a traditional intervention. For example, a couple of studies [37, 38] explored how picture exchanging communication system (PECS) could be implemented via virtual reality. PECS has been known as an effective way to teach functional communications with an initial focus on spontaneous communication in children with ASD. It would be clinically important data if a study shows that PECS via virtual reality is significantly effective as compared with the traditional way. In spite of its significance, it is unfortunate to identify that these studies remained as anecdotic case studies without any follow-up data to support their effectiveness.

Another issue is the quality of simulation technology used in the previous research. Since Ivan Sutherland developed his concept of a virtual world in 1965 [39], major developments in virtual reality have been made in the entertainment and commercial industries. These virtual worlds provide a realistic environment. Virtual reality was also adopted in the areas of fine arts, archeology, and architecture in order to provide spectators with rich sensory experiences. Virtual reality is now available in our daily life. Street View by Google, one of the virtual reality systems that is the most familiar to us, shows realistic panoramic views of roads and buildings with a stereoscopic 3-D mode available to users. While virtual reality is greatly developed in these areas, the quality of simulated images that appeared in a number of papers of speech-language intervention seems somewhat flimsy and unrealistic. It is not certain whether such quality of simulation brings equally

effective treatment effects compared with traditional face-to-face methods.

Finally, there is significant lack of linkage from research to current practice. Although previous studies have successfully inspired practitioners and encouraged them to implement virtual reality in their clinical practice, it is not certain for them where to start and how to start virtual reality since current existing virtual reality is mostly research laboratory prototypes. Research evidence still remains without application to real clinical practice. For example, Cole, Wise, and Vuuren [40], who developed a virtual therapist for adults with aphasia, implemented a similar program for children, reporting that the virtual therapist called Marni successfully behaves like a sensitive and effective reading teacher for children in regular classrooms. Given the successful outcomes with a larger number of normally developing children, it is unfortunate to find no application and follow-up studies with children with language impairments.

Potential Benefits of Virtual Therapy in Healthcare Services

Several potential benefits and usefulness of virtual therapy have been proposed for both service recipients and service providers. First, virtual therapy brings interactivity and motivation to patients [41]. Repetition may be necessary in an actual therapy situation that may reduce patient's motivation. Virtual therapy may overcome the repetitive nature of rehabilitation by providing patients with fun and interesting opportunities to exercise. During intervention, virtual therapy may help to reduce feelings of embarrassment when a breakdown takes place in real-world communication [20••].

Virtual therapy is also considered useful to clients or patients' improvement and generalization in that virtual therapy may effectively control the number of factors affecting patients at each stage of treatment procedures [42]. In other words, during the early phase of treatment, a clinician may reduce the number of distractions in order to improve learning, but the number of distractions can be added to a later phase of intervention to promote generalization. Holden argues that virtual therapy may be superior to real-world practice. Second, virtual therapy may be useful because it allows for immediate feedback on a patient's learning process while immediate feedback is not always plausible in real-world settings [43].

Virtual therapy allows patients to access a program whenever they need to which may lead to an increase in the number of training hours, and consequently a significant improvement in the rehabilitation process. In particular, when virtual therapy is provided via telehealth, the benefits may be maximized. Provider shortages in rural America have been well reported [44]. Also, virtual therapy may help patients improve their outcomes because virtual therapy provides patients with an

opportunity to practice what the clinician directed before they implemented the solution in the real world. For example, patients with dysphagia may be able to practice how to swallow a variety of food consistently without the intake of a real food.

Virtual therapy is also considered useful for training healthcare professionals or students. In medical practice and education, virtual therapy has been used to provide real-time hands-on experience to medical students. Students can have ample experience in a more safe and controlled setting without presenting a risk to patients. In the speech-language pathology profession, students often have limited experience with patients with swallowing difficulties. Also, students may not have ample experience working with low incidence disabilities. Virtual therapy may be beneficial to these students so that they have sufficient experience before applying the skills learned to a real-life case.

Conclusion

This paper reviewed current available research showing the effectiveness of virtual speech-language intervention in children and adults with communication disorders. Although current evidence is somewhat limited, future studies should be conducted with a rigorous experimental design such as randomized controlled design as well as advanced technology.

New technology is being developed every year, and our lives rely on technology more and more. Instead of ignoring it, we need to adopt technology to take advantage of it. However, there are several challenges for this to take place. One challenge would be the equipment cost. Although therapeutic and educational equipment costs have dropped significantly in recent years, current prices are still prohibitive for clinics and schools. Furthermore, more and more virtual therapy is provided via telehealth. An inadequate communication infrastructure may make practitioners hesitant to adopt virtual therapy. Finally, it is not certain whether existing virtual therapy can accommodate specialized individual needs when needed. It is particularly important to clinical practitioners because each patient needs a customized treatment plan. Thus, the development of more user-friendly or easily accommodated virtual therapy programs is highly desired in the future. Without overcoming these limitations, pervasive application of virtual therapy remains uncertain.

Compliance with Ethical Standards

Conflict of Interest The author declares no conflicts of interest relevant to this manuscript.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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